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SLIDE TWO: The past decade has brought major changes to the field of emergency response planning, driven mainly by an increase in incidents requiring widespread communication among stakeholders in the United States. According to the FBI, between the years of 2020 and 2024 and the previous five years, a <SLIDE ADVANCE> 70% increase in active shooter events was recorded in the United States, leading to over <SLIDE ADVANCE> 4,000 victims. Of these mass shootings, more than <SLIDE ADVANCE> 50% were recorded within workplaces and schools.

SLIDE THREE: In 2023, Leander Independent School District passed a bond package that allocated \$10 million for security improvements to its campuses. However, not every school, church, or business has these same resources available to set aside such a large amount to fund security improvements like those proposed by Leander ISD, which could leave the people operating within at risk. As a member of a school widely regarded by interviewees as having a very open campus, I believe that organizations should not be deterred from making security improvements and adopting life-saving technology by implementation costs, and that existing security solutions do not effectively meet the needs of widespread, distributed campus communities like our own.

SLIDE FOUR: To address these issues, the development of this device followed the phases of the Engineering Design Process, including Identify, Describe, Generate, Select, and Embody.

<SLIDE ADVANCE> In the Identify phase, the growing need for communication systems that provide real-time insight into emergencies across large and distributed campuses was established through research and analysis of emergency events.

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SLIDE FIVE: First responders require as much information as possible to plan their strategy and operations, relying on the timeliness and accuracy of the information they receive to make informed decisions that carry the weight of life and death. This flow of timely and accurate information is known as situational awareness, or one's ability to perceive, interpret, and predict dynamic crisis conditions, using environmental knowledge and information about an emergency event. According to Kedia, "Timely, accurate, and complete [situational awareness] is a key enabler of successful disaster response when the situation is changing rapidly, resources are limited, and different agencies must coordinate their activities." With the assurance of high-quality, reliable communications systems during an emergency, first responders and other involved stakeholders can respond most effectively to an event, even amid a variety of unknowns.

SLIDE SIX: Many emergencies require the deployment and use of Emergency Communication Systems (ECS) that are capable of continuously transmitting, processing, and receiving data to enable situational awareness. However, many existing systems on the market fail to deliver the high-quality, uninterrupted communication required in emergencies, with data transfer occurring in simplex, where only one endpoint can send or receive data at a time, leading to missed messages or "stepped-on" voice transmissions. Because effective disaster response requires the rapid dissemination of accurate information across multiple agencies, limitations in infrastructure, communications, data analysis, and interoperability present within these systems can hinder effective decision-making and limit real-time situational awareness.

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SLIDE SEVEN: The transmission of rapid, accurate, and shared information occurs through Internet of Emergency Services (IoES) systems, which – much like the Internet of Things (or “IoT”) devices present within our everyday lives – provide real-time data collection, overcoming delays associated with more manual systems, such as the use of two-way radio. According to (Dah-Mah-Cevice-Eous), “[IoES] technology can help automate and streamline emergency response processes . . . providing real-time data and insights that can enable faster and more effective responses.”

SLIDE EIGHT: By enhancing emergency response systems, the biblical values of protecting life and serving others are upheld. Genesis 1:27 and Psalm 139:13-14 state that “God created mankind in his own image,” and that each individual is “fearfully and wonderfully made.” As such, the implementation of technologies that improve communication and awareness in emergencies not only enables a rapid response to events but also honors the image of God within every person.

SLIDE NINE: This demonstrates the need for this project: Enabling situational awareness during emergency events through the utilization of inter-building nodes and a mobile application

SLIDE TEN: Throughout the rest of this presentation, the Engineering Design Process will be used to justify the two components that comprise the design: an intra-room device (or “node”) placed within a building to act as a beacon for user mobile devices and serve as an offline communications network, and a mobile application to interface with the beacon.

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SLIDE ELEVEN: In the Describe Phase of the Engineering Design Process, <SLIDE ADVANCE> I examined many existing approaches to maintaining situational awareness during emergencies, demonstrated the need for a solution independent of building infrastructure, and identified the requirements of stakeholders involved in emergency response and management.

SLIDE TWELVE: Historical FBI data from 2000-2013 recorded an average active shooter event duration of 12 minutes, while more recent incidents, such as the Perry and Apalachee shootings, have fallen within a shorter 4-to-6-minute range, highlighting shrinking response windows for first responders. Additionally, mass shooting events are statistically more likely to occur in workplaces and schools, which comprise of 28% and 22% of all recorded events, respectively. While legislative requirements such as Alyssa's Law mandate silent alarms in Texas' public schools, the timeline of recent incidents shows that these are often less effective than a simple 911 call, as they cannot communicate rich data, reinforcing the need for a more adaptive and distributed solution that keeps pace with the dynamic nature of an emergency event.

SLIDE THIRTEEN: The first existing solution analyzed was a patent for an Integrated Emergency Data Platform that combines data sources such as geographic datasets, phone records, and traffic networks into one single interactive graphical display for emergency communication centers. While systems like these can reduce response times by sharing information such as floor plans, video feeds, and access controls with responders, their implementation relies on pre-existing infrastructure, including stable power and internet access and building-embedded sensors, which can fail after physical damage occurs. State-level stakeholder interviews highlighted that the overwhelming volume of data these platforms produce can make decision-making difficult, especially when responders need to leave their command-and-control workstations and operate remotely in the field.

SLIDE FOURTEEN: The second existing solution that was analyzed was an automated crisis incident response system for IoT networks that detects events, such as gunshots, and triggers a response plan by controlling connected IoT devices across a facility. These systems can automate emergency procedures, standardize crisis response, and minimize confusion. However, these solutions rely on cloud-based communication and assume continuous internet access. As a result, while these systems are effective under normal conditions, their operation is significantly degraded when connectivity is lost, which may be exactly when stakeholders rely on the system the most.

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SLIDE FIFTEEN: To identify the specific needs and requirements of this project, interviews were conducted with business, local, and state-level emergency response coordinators and personnel. Across all interviews, stakeholders highlighted the need for a fast, reliable system. The highest-priority areas included direct, delay-free communication between stakeholders, rapid and accurate information distribution, and systems that remain functional even after power or connectivity failures. Stakeholders also emphasized the importance of a platform that integrates well with existing systems and provides a simple interface for untrained personnel to interact with.

SLIDE SIXTEEN: With the stakeholder needs identified, the project then moved to the Generate Phase of the Engineering Design Process, where potential solutions were brainstormed to fit the established needs.

SLIDE SEVENTEEN: The brainstorming process began with a mind map created with a group of students, which was used to determine requirement categories and potential solutions. Each category was based upon key stakeholder needs, including ease of use, speed, multi-modal compatibility, reliability, and accuracy.

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SLIDE EIGHTEEN: After completing the mind map, our group moved on to concept sketches to generate ideas that met the requirements specified in the mind map. Each concept incorporated a sensing element, connected to a communications system that enables the real-time observation of emergency events, through a drone-based solution in Concept A and a robotics-based solution in Concept B. Unfortunately, after further analysis, neither solution met the stakeholders' needs for information distribution, low cost, or easy integration with a building's infrastructure. As such, the Generate Phase was revisited, and concepts in the Mind Map were reevaluated to determine the proposed solution, which enabled the localization of key personnel within a building using inter-room Bluetooth Low Energy nodes.

SLIDE NINETEEN: The project then moved into the <SLIDE ADVANCE> Select Phase, where portions of the final concept were evaluated, their strengths and weaknesses identified, and a final design selected for testing. This phase involved concept analysis using Pugh Charts and risk analysis using a Failure Modes and Effects Analysis (FMEA) table.

SLIDE TWENTY: The first Pugh Chart was used to identify what the user-facing interface of the system would look like, with the selected datum being an existing push-to-talk two-way radio system used by many security personnel interviewed during the Describe Phase. Of the options, including a Voice over IP radio, body pack, and voice-activated headset, the smartphone application most clearly met each identified need, scoring a net positive six. This is due to its multimodal nature, widespread presence, and user familiarity, making it highly adaptable across a variety of deployment environments. The Voice over IP radio had the advantage of enabling duplex communication, meaning audio could be sent and received simultaneously; however, issues arose with the added hardware cost and signal coverage, placing it below the smartphone application. Two other interfaces discovered during the ideation process were both wearables, one in the form-factor of a headset and the other in the form of a body-worn sensor pack. Each was deemed too difficult to interact with physically and would increase deployment complexity in a variable environment.

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SLIDE TWENTY-ONE: The second Pugh Chart was used to determine how Internet of Emergency Services technology could be incorporated into the smartphone application to provide actionable insights for stakeholders and fulfill the need for actionable situational awareness. The selected datum for this Pugh chart was an existing solution that relied on sharing emergency procedures via paper binders in each room on campus. Of the identified concepts, two additions to the smartphone application were prioritized based on the Pugh Chart: the dynamic rule-based emergency guideline system and the Bluetooth Low Energy room localization system, both of which scored a net positive five against the datum. The dynamic rule-based emergency guideline system would use a mobile application to push real-time updates to emergency response plans, ensuring communication among all parties involved in emergency response and reducing cognitive load. The Bluetooth Low Energy room localization system would allow stakeholders to view the locations of key personnel within a building during an emergency, increasing situational awareness and knowledge of unfolding events.

SLIDE TWENTY-TWO: With a solution selected, the project then moved to the Embody Phase, where the concept was designed and tested to identify strengths and weaknesses with iterative testing, and finalized into a design that meets all stakeholder needs.

SLIDE TWENTY-THREE: The first test of the system was designed to assess battery life under simulated degraded infrastructure conditions, ensuring that the stakeholders' need for infrastructure independence is met. The test was conducted on both short-range (BLE Only) and long-range (LoRA) nodes, with BLE beacons broadcast at 5, 15, and 30-second intervals over one hour. The data collected showed a mean activation time of 467 hours for short-range nodes and 110 hours for long-range nodes, exceeding the duration of any realistic emergency event and confirming that the system can operate independently during crises. The system was tested in both the short- and long-range radio configuration to fulfill the stakeholder requirement for infrastructure independence, allowing nodes to communicate regardless of distance to the nearest node and enabling system customization to the deployed environment.

SLIDE TWENTY-FOUR: The second test of the system was designed to determine the accuracy of BLE localization in the mobile application. The test was conducted by changing both the nRF52 BLE output power and the mobile device's distance from the node, then recording the change in Received Signal Strength Indicator, or RSSI, at each setting. Tests were conducted at <SLIDE ADVANCE> +/- 4 decibel-milliwatts, and <SLIDE ADVANCE> 0 decibel-milliwatts of output power. This established a signal profile using the log-distance path-loss model, thereby improving the localization algorithm's accuracy in determining a device's position within a building. The collected data showed a direct correlation between real distance and calculated distance as RSSI varied, confirming the device's ability to determine its room-level position regardless of beacon transmission power, using a well-tuned path-loss model.

SLIDE TWENTY-FIVE: The third test of the system was designed to determine the maximum effective indoor range of the LoRa radios, ensuring that mesh-based communication between long-range bridge nodes is possible. Fifty packets were sent between two LoRa nodes at measured distances of 10, 25, and 50 feet, both on a single floor and between floors of a multi-story building, and inside portable buildings to account for differences in building material characteristics. The data collected showed a 100% success rate at short distances in the main building, with signal degradation increasing with distance, floor separation, and denser construction. <SLIDE ADVANCE> These results allow us to set the recommended density of one long-range bridge node per floor or per cluster of distributed buildings, depending on surrounding signal characteristics.

SLIDE TWENTY-SIX: The final test of the system verified communication between the mobile application and the mesh network of nodes. <SLIDE ADVANCE> An emergency event was triggered from the application menu, <SLIDE ADVANCE> the alert was propagated through the network, <SLIDE ADVANCE> and role-based tasks were assigned to connected users. Backend data logs <SLIDE ADVANCE> confirmed that both the event-creation and task-assignment payloads were transmitted and received correctly over the LoRa mesh and the internet, verifying the system's online and offline communication capabilities.

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SLIDE TWENTY-SEVEN: The final design of the room node is built around several components. Each node uses an nRF52 System-on-Chip running the Zephyr Real-Time Operating System, chosen for its low power and robust Bluetooth Low Energy stack. BLE advertisements are broadcast on the 2.4 GHz band to enable room-level localization on stakeholder mobile devices. Long-range nodes include a LoRa radio operating in the 900 MHz band to transfer data across distributed buildings where BLE alone is insufficient. Each node also contains a 3.7-volt 1200-milliamp-hour Lithium-Ion Polymer backup battery and is primarily powered through Power over Ethernet during normal operation. This results in two node configurations, short-range and long-range, that can be tailored to the specific environment in which they are deployed.

SLIDE TWENTY-EIGHT: The final design of the mobile application determines a user's position relative to a building during an emergency event using Bluetooth Low Energy broadcasts from room nodes, and updates this location through both the internet and the mesh network of nodes created during an emergency event to ensure redundancy. During an event, each user's location is displayed on the building's floor plan, and only users with configured access levels can view this data. Additionally, emergency alerts are sent to every enrolled device, and the dynamic playbook system automatically assigns role-specific tasks based on the selected emergency response.

SLIDE TWENTY-NINE: The images shown represent the triggering mechanism for an emergency event, which can be done by any user of the mobile application. Different types of emergencies are defined to ensure full playbook coverage, and each emergency type can have multiple playbooks within it – as shown by the different building locations for the two fire emergency playbooks.

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SLIDE THIRTY: These tasks are then assigned to users in an easy-to-understand to-do list format, helping reduce the application's cognitive load during an emergency event and serving as a central store of actionable items.

SLIDE THIRTY-ONE: The application is configured to only allow users with the correct access level to view the current status of playbooks and user locations during an emergency event. In the case shown, only users with the Admin or Security designation may access the screen shown in the image, which contains the playbook tracker and emergency resolution button.

SLIDE THIRTY-TWO: This video shows the process of creating a new playbook in the system.
[EXPLAIN AS ITEMS COME ON SCREEN]

SLIDE THIRTY-THREE: This video shows the process for triggering an emergency event and checking the status of users' playbooks and location updates. [EXPLAIN AS ITEMS COME ON SCREEN]

SLIDE THIRTY-FOUR: One counterargument to the proposed solution is the reliance on stakeholders' personal mobile devices as the system's interface. Depending on personal devices introduces risk, as users may have their phones powered off, silenced, or packed away, and in an emergency where seconds matter, even a small number of devices failing could hinder situational awareness. However, the decision to use a mobile application was made because it offers lower latency, reduced cognitive load, support for multimodal data, and greater security. The ubiquity of smartphones reduces the need for additional hardware and user training, directly addressing the high installation costs this project sought to address. FMEA analysis also identified countermeasures for this scenario, including dedicated response devices stationed in critical zones and organizational policies governing device usage during emergencies. Additionally, the mobile application is programmed to override mobile phone settings, such as Do Not Disturb, providing an additional layer of security.

SLIDE THIRTY-FIVE: A second counterargument is that this system is not truly "infrastructure-free," as it relies on Power over Ethernet being available within the building during normal day-to-day operations. Creating an ultra-low-power microcontroller using off-the-shelf components is difficult because manufacturers typically do not prioritize power efficiency when laying out their PCBs. PoE provides a suitable alternative for large, distributed campuses that already have wireless networking infrastructure in place, and the system retains full independent operation during emergencies by using its built-in backup battery when infrastructure fails.

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SLIDE THIRTY-SIX: The next step in this project's design would be to create and manufacture a fully custom PCB that integrates all electronic components into a single package. Consolidating all components onto a single custom-designed board would reduce per-unit cost by eliminating Commercial Off-the-Shelf profit margins, improve power efficiency by leveraging ultra-low-power features, and allow for smaller physical packaging. While custom PCB manufacturing was explored for this project, limitations in single-board ordering, component availability, and design knowledge gaps made it infeasible.

SLIDE THIRTY-SEVEN: In conclusion, the frequency of emergency incidents has raised awareness of how organizations maintain situational awareness and coordinate response during rapidly evolving events. This emergency localization system provides low-cost room-level positioning via BLE beacons, mesh-based communication independent of building infrastructure, and role-specific playbooks that reduce cognitive load during emergencies. By lowering the barrier to implementation, this system enables schools, churches, and other organizations to deploy life-saving emergency communication technology without overhead that could otherwise dissuade investment.

SLIDE THIRTY-EIGHT: Thank you. I will now take any questions.